

In your previous classes you might have studied about telescope and microscope. Telescope helps to see objects at distance whereas microscope enlarges the objects thousand times of their size. There might be few students in your class who wear glasses to read books and blackboard. Have you ever wondered why this is so? When you see through the microscope or a telescope you might have seen a transparent glass in the eye piece. These are **Lenses**.

Look at the pictures given below and answer the questions.

- Why are these children using telescope and microscope?
- How a child can see distant objects through telescope?
- How a child can see tiny organisms from the microscope?

In this Chapter you will learn about:

- Lenses
- Types of Lenses (Converging and Diverging Lenses)
- Image Formation by Ray Diagram
- Image Formation in Simple Camera and Human Eye
- Uses of Lenses

All the students will be able to:

- Define lens.
- Differentiate between the different types of lenses.
- Describe the image formation using a lens by ray diagram.
- Compare and contrast the working of a human eye with the lens camera.
- Explain how eye focuses by altering the thickness of the eye lens.
- Investigate how eyes get used to darkness after some time.
- Explain how lenses are used to correct short sightedness and long sightedness.
- Identify the types of lenses used for various purposes in daily life.



Fig.10.1: A Child using Microscope



Fig.10.2: A Child using Telescope

LENSES

- ✓ Define lens

A lens is a transparent curved piece of glass or plastic designed to refract light in a specific way. Each surface of lens is a part of a sphere. You must have studied in your previous classes that when a light ray moves from one medium (like air) to another medium (like glass), it bends. This is called refraction. By using refraction, lenses can bend multiple light rays. Most of the lenses we use in everyday life are designed to bend light rays to a specific focal point where objects get clearer.

TYPES OF LENSES (CONVERGING AND DIVERGING LENSES)

- ✓ Differentiate between the different types of lenses.

There are two main types of lenses. Convex (converging lens) and Concave (diverging lens).

1. Convex or Converging Lens

Converging lenses can be identified by their shape; they are relatively thick across their middle and thin at their upper and lower edges. A converging lens is curved outwards on both sides. It causes the light rays to bend to a specific focal point, hence, converges rays at a specific point making a real image on screen.

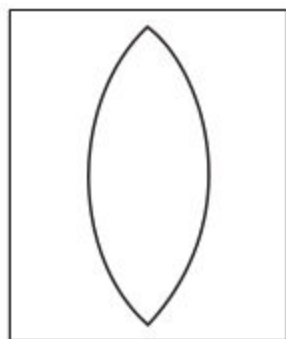


Fig.10.3: Convex Lens

2. Concave or Diverging Lens

Diverging lenses can also be identified by their shape; they are relatively thin across their middle and thick at their upper and lower edges. A diverging lens is curved inwards on both sides. Diverging lenses refract the parallel rays of light so that they spread apart from one another. This means that they form a virtual image virtual image cannot be projected on a screen.

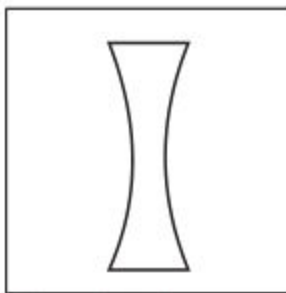


Fig.10.4: Concave Lens

IMAGE FORMATION BY RAY DIAGRAM

✓ Describe the image formation using a lens by ray diagram

The language of a lens: There are certain terms which we use relative to the lens.

- 1. Optical Centre:** The centre of a lens is called optical centre.
- 2. Centre of Curvature:** The centre of a sphere is called centre of curvature of which lens is a part.
- 3. Vertical Axis:** A line that bisects lens into two halves.
- 4. Principal Axis:** The line passes through the optical centre and centre of curvature of the faces of the lens is called principal axis.
- 5. Principal focus (F) of a Lens:**
 - **Convex lens:** In case of convex lens the light rays parallel to the principal axis after refraction through the lens meet at a point. This point is called principal focus (F). As the light rays actually meet at the focus point after refraction through the lens, so the focus point is real. Since convex lens converges light rays at principal focus that is why it is called converging lens. Because of this property convex lens makes real images on the screen placed on the other side of the lens.
 - **Concave lens:** In case of concave lens, light rays parallel to the principal axis after passing through the lens bend in such a way that they do not meet at a one point. They diverge out and appear to be coming from one point which is called principal focus. The principal focus of a concave lens is virtual. That is image cannot be taken on the screen.

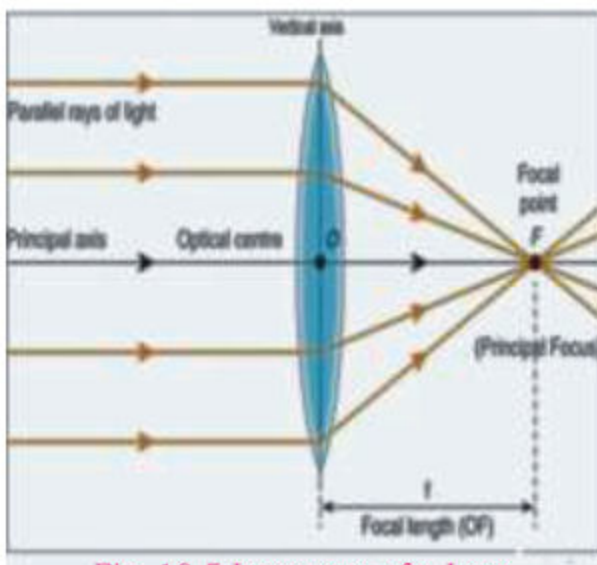


Fig. 10.5 Language of a Lens

6. Focal length of a Lens (f):

- **Convex lens:** The distance between the optical centre and the principal focus is called focal length (f). Focal length of a convex lens is taken as positive.
- **Concave lens:** The distance between the optical centre and the principal focus is called focal length (f). Focal length of a concave lens is taken as negative.

Image formation by convex lens using ray diagram.

Following steps should be followed for drawing ray diagram.

1. Draw a double convex lens and pass the line that is principal axis through the optical centre of the lens.
2. Draw an object to the left side of the lens represented by an arrow
3. Draw one ray parallel to the principal axis and bend it from the centre of the lens passes through the focus.
4. Draw another ray from the object passes through the centre of the lens.
5. The point at which these rays intersect each other after refraction is the top of the image.

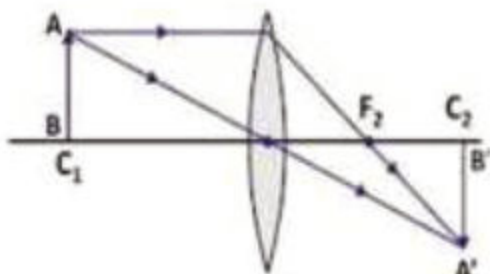


Fig.10.6: Ray Diagram

Image formation by a concave lens using ray diagram.

In case of concave lens, rays diverge out and do not meet on the other side of the lens. The image will always be virtual, upright and smaller in size and is formed on the same side of the lens on extending the rays backward.

1. When the object is at infinity

When the object is at infinity, the rays are parallel and strike the lens and get diverged. Extend the rays backward and pass them from the focus. An erect, virtual and diminished image is formed at the focus on the same side as the object.

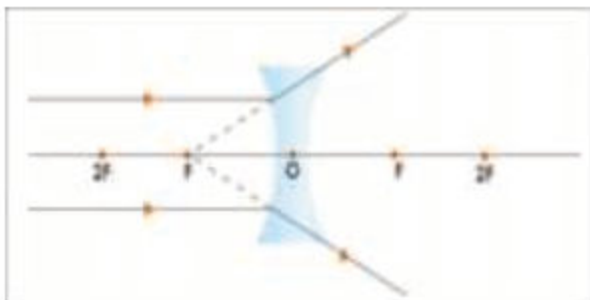


Fig.10.7 : Object is at Infinity

2. When the object is placed in front of F

Draw a ray parallel to the principal axis and diverge it from the lens. Draw a second ray and pass it from the optical centre. Extend the first ray and pass from the focus. At the point where two rays meet, will the image form. When the object is placed at any position between O and infinity, an erect, virtual and diminished image is formed between O and F.

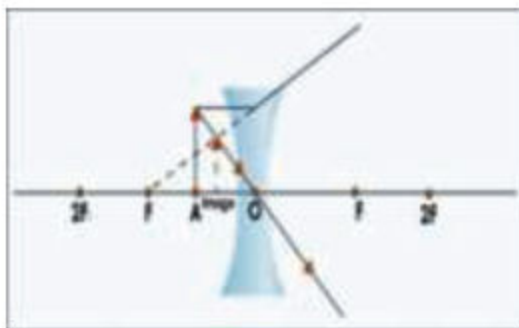


Fig.10.8: Object is placed in front of F

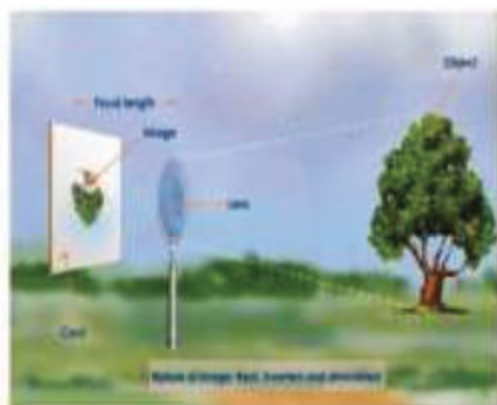
Activity 10.1:

Image formation by a convex lens.

What you need: Magnifying glass, a blank wall outside of your class where you get light.

What you do:

1. Stand in front of the blank wall where you get light.
2. Hold a magnifying lens in your hand and try to take image of any distant object on the wall e.g. tree.
3. Move the lens back and forth to take the clearer sharp image.
4. Note down the characteristics of the image in the observation table.



What you observe:

	Size of image: larger or smaller	Nature of image: real/virtual/erect/u pside down
When we moved lens towards the object		
When we moved lens away from the object		

Activity questions:

1. Which type of image is formed by the magnifying glass on the wall?
2. Can you do this activity by a concave lens? Explain.

Activity 10.2:

Exploring lenses

What you need:

Science notebook, convex lenses, concave lenses, flashlights, white paper, and note book.

What you do:

1. Look closely at the convex lens and note down the surface and shape of the lens in the observation table.
2. Look closely at the concave lens and note down the surface and shape of the lens in the observation table.
3. Look through the convex lens at the pages of your book, your hands, a hair, and other things. Note down how does the convex lens make things look.
4. Now look through the concave lens at the pages of your book, your hands, a hair, and other things. Note down how does the concave lens make things look.
5. Lit a flashlight through the convex lens onto a piece of white paper and then note down in which direction does the convex lens bend the light.
6. Lit a flashlight through the concave lens onto a piece of white paper and then note down in which direction does the concave lens bend the light.
7. Now keep both the lenses in a straight line, faces towards each other and shine a flashlight. Draw pictures of bending of light rays in your science notebook.

What you observe:

Lenses	Shape and surface	Lens makes things look	Direction: bending of light
Convex			
Concave			

Activity questions:

1. How are both lenses alike?
2. How are both lenses different?
3. What happened when you use both lenses at the same time?
4. Can you use both lenses to make distant things appear closer?
5. Which lens converges light rays?
6. Which lens diverges light rays?

IMAGE FORMATION IN SIMPLE CAMERA AND HUMAN EYE

- ✓ Compare and contrast the working of a human eye with the lens camera.
- ✓ Explain how eye focuses by altering the thickness of the eye lens.
- ✓ Investigate how eyes get used to darkness after some time.

Camera:

The main parts of the camera involved in the process are: the camera body, the camera shutter, the camera lens, the lens aperture, and the camera's image sensor or the film. The camera's LCD screen is for previewing and then viewing the captured image.

The camera body is a light proof box. The controls for exposure settings and other effects are located on the camera body. The camera shutter and the image sensor are located inside the camera body. Depending on the type of camera, the camera lens may be permanently attached to the camera (fixed lens) or removable (interchangeable lens). When the button is pressed to release the camera's shutter, it opens and any light flowing into the camera lens is directed through the lens

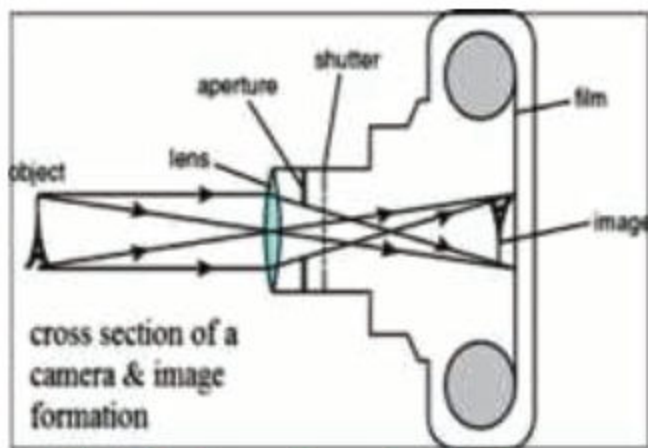


Fig.10.9: Image formation in Camera

Teacher Note:

Make groups of 5-6 students each. Give one convex lens, one concave lens and one flash light to each group. Assist students wherever required.

aperture and the open shutter to the camera's image sensor. The picture is obtained by developing the image on the film or by getting print from the computer.

Human Eye:

The given figure 10.10 shows different parts of human eye.

Light from an object passes through the cornea which is a transparent dome like structure covering the iris. The light rays are refracted by the cornea onto the lens. The light rays are refracted second time while passing through the lens and focused onto the retina – the light sensitive part of the eye. The image formed on the retina is inverted (upside down) and real (the light rays travel through the image). The image is interpreted the right way up by the brain which is connected to the eye via the optic nerve.

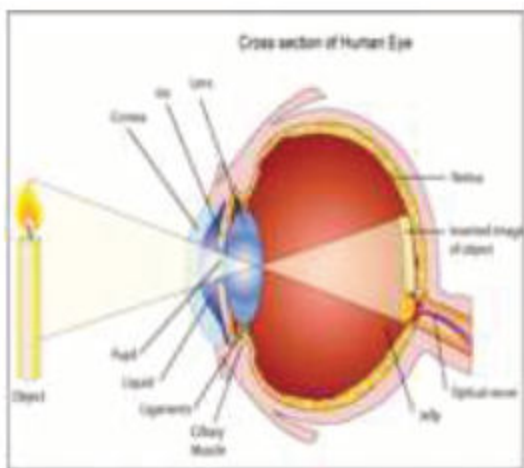


Fig.10.10:Cross-section of Human Eye

Camera and Human Eye: Similarities

I) Parts:

a) Cornea of an eye and front lens of the camera

A convex lens is mounted on the front side of a camera. Your cornea behaves much like the front lens of a camera. Together with the lens, which is behind the iris, they are the eye's focusing elements. The cornea takes widely diverging rays of light and bends them through the pupil, the round opening in the central portion of the coloured iris.

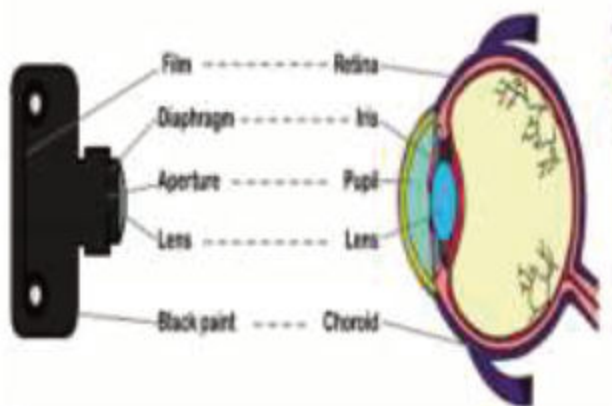


Fig 10.11: Camera Versus Human Eye

b) Iris and pupil of an eye and aperture of a camera

There is an opening in the diaphragm behind the lens which controls the light. Your iris and pupil act like the aperture of a camera. The iris is a muscle which, when contracted, covers all but a small central portion of the lens allows adjustable control of the quantity of light entering the eye so that the eye can work well in a wide range of viewing conditions, from dim to very bright light.

c) Retina of an eye and film of a camera.

A real and inverted image is formed on the film of a camera. Our Retina is the sensory layer that lines the very back of our eyes. It acts very much like the imaging sensor chip in a digital camera or like a film in a film camera where image is formed. The retina has numerous photoreceptor nerve cells that help change the light rays into electrical impulses and send them through the optic nerve to the brain where an image of what we see is finally received and perceived. Because of this reception and perception function retina is perhaps the most important component of our eyes.

II) Working

- 1. Image focusing:** Human and camera lenses both focus an inverted image onto light-sensitive surface. In case of a camera, it is focused onto film or a sensor chip. In your eyes, the light-sensitive surface is the retina on the inside of your eyeball.
- 2. Light adjustment:** Both the eye and a camera can adjust quantity of light entering. In a camera, it is done with the aperture control built behind the lens; while in your eye it is done by having a larger or smaller iris.

Camera and Human Eyes: Differences

Your eyes work in harmony with your brain to create the images you perceive. Your eyes are adjusting the focus by bending the light through the lens in your eyeballs

DO YOU KNOW?

The retina contains about 100 million rods and cones. If you consider each of them as a pixel then the human eye has more than a Cannon camera.

and translating light into an electrical impulse your brain can process. From there onwards, it is all about your brain. It is continuously readjusting its colour balance according to the lighting context. In other words, your eyes know what must be seen as red or white or black etc. In a camera it measures the light that hits a series of sensors, but the sensor and the signals recorded need to be adjusted to suit the color temperature of the light illuminating the scene.

a) Lens focus: In camera, the lens moves closer or further from the film to focus. In your eyes, the lens changes shape to focus. The muscles in your eyes change the actual shape of the lens inside your eyes.

b) Sensitivity to light: A film in a camera is uniformly sensitive to light. The human retina is not. Therefore, with respect to quality of image and capturing power, our eyes have a greater sensitivity in dark locations than a typical camera.

How Eye Focuses Light

The human eye is a sense organ adapted to allow vision by reacting to light. The cornea and the crystalline lens are both important for the eye to focus light.

The crystalline lens and accommodation

The ciliary muscle is a circular ring of muscle that attaches all the way around the lens. This ciliary muscle can change the shape of the crystalline lens by stretching it at the edges. When you are looking at a near object, the lens needs to become more rounded at the central surface to focus the light rays. The eye can alter the shape and

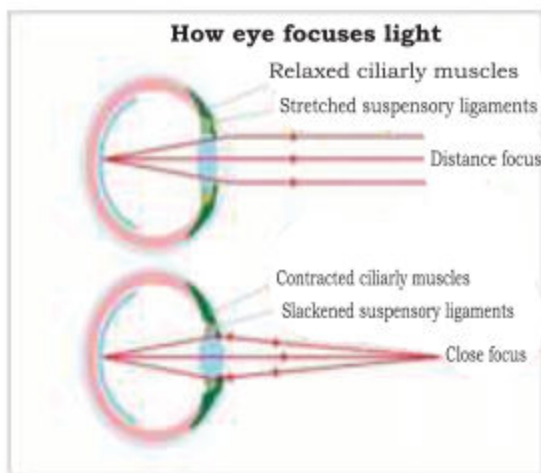


Fig.10.12: How Eye Focuses Light

curvature of the lens to adjust the degree of refraction. This is called accommodation. It allows light to be focused onto the retina from near or distant objects. Accommodation is achieved by the contraction or relaxation of the ciliary muscles, which slacken or stretch the suspensory ligaments. The table summarizes how accommodation works.

Object	Ciliary muscles	Suspensory ligaments	Muscle tension on the lens	Lens shape
Near	Contracted	Slackened	Low	Thick and more curved
Distant	Relaxed	Stretched	High	Thin and less curved

Suppose you are playing a football in a playground in a bright sunny day. After first half you go to the rest room to have some soft drink where the light is turned off. You will notice that you are having difficulty in seeing things but after some time you are able to see little and later you can see things clearly. Similarly, when you go out in the sunlight from a dark room it is very difficult to keep your eyes open. Think, why does it take your eyes several minutes to adjust to darkness or bright light? Our eyes are amazingly versatile, we can see perfectly well in a vast variety of light conditions. It is our visual system that is able to adapt to brilliant sunshine and then to darkness when required. What makes this possible? The answer lies in two distinct parts of the eye – the pupil and the retina.

DO YOU KNOW?

You can only see three colors; red, blue, and green. All other colors are created through the combination of these three colors. The light, which is colorless, reflects off objects and is then perceived by the eye.

The pupil

The pupil can expand and contract depending on the amount of light available. The pupillary light reflex controls the diameter of the pupil, when the light intensity is greater, the pupil becomes smaller and allows less light in. When it is dark, it instructs the pupil to become larger to enable more light to enter the retina.

Retina

Retina contains rods and cone cells. Rod cells are responsible for vision in dim light or in darkness. Cone cells are responsible for vision in bright light and coloured vision. Rhodopsin is a chemical found in the rods. When you go in the bright light, it breaks down into two molecules, the retinal and the opsin. When you go back in the darkness or in a very dim light, these two molecules recombine into rhodopsin molecule. The re-combination of two molecules is slow, that is why you can see in the darkness after some time.

DO YOU KNOW?

The retinal used in the eye is derived from vitamin A. If a person's diet is low in vitamin A, there is not enough retinal in the rods and therefore not enough rhodopsin. People who lack vitamin A often suffer from night blindness, they cannot see in the dark.

USES OF LENSES

- ✓ Explain how lenses are used to correct shortsightedness and long sightedness.
- ✓ Identify the types of lenses used for various purposes in daily life.

The shape of the eye is very important in ensuring the objects we see are in focus. A person with normal vision can focus clearly on objects both near and far, this is because light from the object is exactly focused onto the retina at the focal point. However, for some people focusing on objects far away or close can result in a blurred image forming. These defects in vision are referred to as long-sightedness and shortsightedness.

Long-sightedness

A person who is long sighted can focus clearly on distant objects

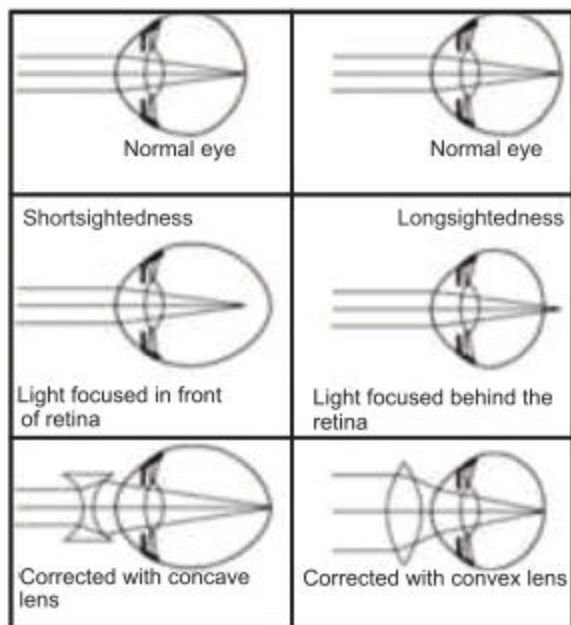


Fig.10.13: Long-sightedness & Short-sightedness

but cannot focus on near objects. This is because the eyeball is too short. Light from near objects is focused at a point behind the retina resulting in a blurred image. Fig. 10.13

This defect can be corrected by wearing a convex (converging) spectacle lens. The rays of light from a near object are converged before entering the eye so that the cornea and eye lens can direct the focal point onto the retina.

Short-sightedness

A person who is short sighted can focus clearly on near objects but cannot focus on distant objects. This is because the eyeball is too long. Light from distant objects is focused at a point in front of the retina resulting in a blurred image. This defect can be corrected by wearing a concave (diverging) spectacle lens. The rays of light from a near object are diverged before entering the eye so that the cornea and eye lens can direct the focal point onto the retina.

Lenses exist in a variety of places around us, from the interior of the human eye to the inner workings of computer memory systems. The following are major uses of lenses in daily life:

1. **Magnifying glass:** Convex lens is used as magnifying glasses to magnify objects.
2. **Eye glasses:** Convex lens is used for the correction of long-sightedness and concave lens is used for the correction of shortsightedness.
3. **Contact lenses:** A contact lens, or simply contact, is a thin lens placed directly on the surface of the eye. Like eye-glasses, contact lenses help to correct refractive errors.
4. **Different optical instruments:** In binoculars, monocular telescopes, microscopes, cameras and projectors different types of lenses are used.
5. **Flashlight:** Concave lenses are used in flashlights to magnify the light produced by the bulb. The light falls on the concave side of the lens, and the rays diverge on the other side, thereby increasing the apparent radius of the light

source and providing a wider beam.

6. **Lasers:** Small concave lenses are used in various types of medical equipment, scanners and CD players to widen the laser beams.
7. **Peepholes:** Door viewers, or peepholes are small security devices that provide a wide view of objects and environments outside doors or walls. The view is created using one or more concave lenses inside the device which minimizes the proportions of specific objects and gives a wide overview of an entire area.



Fig. 10.14 Uses of Lenses

SUMMARY

- A lens is an optic piece of curved transparent glass that refracts light in a specific way.
- There are two types of lenses: convex and concave.
- Convex lens converges light whereas concave lens diverges light.
- A converging lens is thicker at the middle than its edges whereas a diverging lens is thinner at the centre than its edges.
- The centre of the lens is called optical centre.
- The distance between the optical centre and principal focus is called focal length.
- There are certain terms which we use relative to the lens. i.e. optical center, principal axis, principal focus, focal length, vertical axis.
- Concave lens always forms virtual images.
- There are certain parts of human eye that work like camera. Like cornea and lens, aperture, iris, pupil, retina work like lens aperture and film of a camera.
- Eye focuses by altering the thickness of the eye lens. More curved for the nearer objects; thinner for the far objects. Ciliary muscles and suspensory ligaments help in this action.
- In long-sightedness image is formed at the back of the retina and it is treated by convex lens.
- In shortsightedness image is formed in front of the retina and it is treated by concave lens.
- Magnifying glass, microscope, telescope, projector, vision spectacles are some major uses of lenses.

EXERCISE

1. Choose the correct answer.

- i. A horizontal line that passes through the lens is called:
a) vertical axis. b) principal axis.
c) optical centre. d) focal point.
- ii. The point through which a ray of light passes without changing its path is the:
a) centre of curvature. b) mid point at F and O.
c) principal focus. d) optical centre.
- iii. To obtain sharp image in a camera:
a) lens is moved back & forth.
b) film is moved back & forth.
c) both the lens and the film are moved.
d) neither lens nor film are moved.
- iv. Which of the following helps to change the shape of the lens in the eye?
a) Ciliary muscle. b) Cornea.
c) Retina. d) Iris.
- v. The distance between the optical centre and the principal focus is called:
a) principle axis. b) vertical axis.
c) focal length. d) optical centre.

2. Fill in the blanks.

- i. When the object is at infinity, the rays are parallel and strike the _____ lens and get diverged.
- ii. An erect and diminished image is formed when the object is placed in front of _____ of concave lens.
- iii. A _____ lens is used as a magnifying glass.
- iv. Light from near objects is focused at a point behind the retina in the eye, hence creates defect called _____.
- v. In the eye defect shortsightedness, the eyeball is too _____ and the light from the distant object is focused _____ of the retina.

3. Complete the given table.

S No:	Parts of a camera	Similar part of an eye	Similar function
1.	Aperture		
2.	Lens		
3.	Film or image sensor		

4. Answer the following questions.

- Describe the formation of ray diagram by convex lens.
- Why an image can not be formed on the screen by a concave lens? Explain.
- How is the focal length of a lens affected by its thickness?
- Draw a ray diagram for long-sightedness.
- Which type of lens do you have in your eye?
- Write THREE differences between long-sightedness and short-sightedness.

5. Write similarities and differences of convex and concave lenses in the given Venn diagram.

